First-Order Differential Equations

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2026-10-21

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Notes with worked examples: analytical methods (separable, linear, exact), direction fields, an introduction to numerical methods, and applications.

Contents 1. Definitions and initial value problems 2. Direction fields 3. Separable equations 4. Linear equations (integrating factor) 5. Exact equations 6. Numerical methods: Euler and RK4 (the full treatment is in the Numerical Methods chapter) 7. Applications: Newton's law of cooling, logistic growth 8. An equation with no elementary solution 9. Summary 10. Exercises

```
import numpy as np
import matplotlib.pyplot as plt

plt.rcParams["figure.figsize"] = (7, 4.5)
plt.rcParams["axes.grid"] = True
np.set_printoptions(precision=6, suppress=True)
```

1. Definitions

A **first-order ordinary differential equation (ODE)** relates an unknown function $y(x)$ to its first derivative:

$$\frac{dy}{dx} = f(x, y).$$

An **initial value problem (IVP)** adds a starting condition:

$$\frac{dy}{dx} = f(x, y), \qquad y(x_0) = y_0.$$

- A **general solution** contains an arbitrary constant C (a one-parameter family of curves).
- A **particular solution** is the member of that family that satisfies the initial condition.

Existence and uniqueness (Picard's theorem). If f and $\partial f/\partial y$ are continuous in a rectangle containing (x_0, y_0), the IVP has exactly one solution on some interval around x_0.

Example where uniqueness fails: $y' = 3y^{2/3}$, $y(0) = 0$. Both $y = 0$ and $y = x^3$ are solutions, because $\partial f/\partial y = 2y^{-1/3}$ is not continuous at $y = 0$.

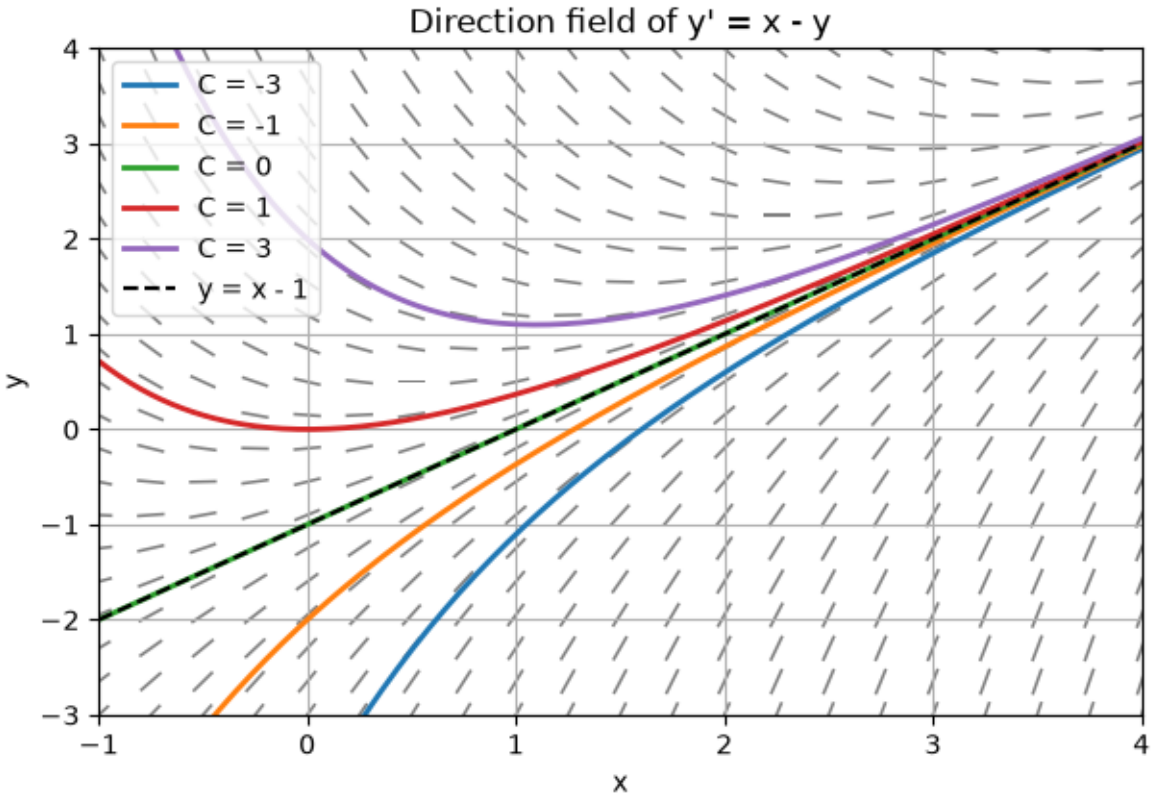
2. Direction fields

At every point (x, y) the equation gives the slope $f(x, y)$ of the solution curve passing through it. Drawing short line segments with these slopes produces a **direction (slope) field**. Solution curves follow the arrows.

Below is the field for $y' = x - y$, with a few solution curves. The exact solution is $y = x - 1 + Ce^{-x}$, so every curve approaches the line $y = x - 1$.

```
def direction_field(f, xlim, ylim, n=21, ax=None):
    # Draw normalised slope segments of y' = f(x, y).
    ax = ax or plt.gca()
    X, Y = np.meshgrid(np.linspace(*xlim, n), np.linspace(*ylim, n))
    S = f(X, Y)
    L = np.sqrt(1 + S**2)
    ax.quiver(X, Y, 1 / L, S / L, angles="xy", color="gray",
              headwidth=0, headlength=0, headaxislength=0, pivot="middle")
    ax.set_xlim(xlim); ax.set_ylim(ylim)
    ax.set_xlabel("x"); ax.set_ylabel("y")
    return ax

f = lambda x, y: x - y
ax = direction_field(f, (-1, 4), (-3, 4))
x = np.linspace(-1, 4, 200)
for C in [-3, -1, 0, 1, 3]:
    ax.plot(x, x - 1 + C * np.exp(-x), lw=2, label=f"C = {C}")
ax.plot(x, x - 1, "k--", label="y = x - 1")
ax.set_title("Direction field of y' = x - y")
ax.legend(loc="upper left");
```



3. Separable equations

An equation is **separable** if it can be written as

$$\frac{dy}{dx} = g(x)\,h(y).$$

Method: separate the variables and integrate both sides:

$$\int \frac{dy}{h(y)} = \int g(x)\,dx + C.$$

Worked example 3.1

Solve $\dfrac{dy}{dx} = xy$, $y(0) = 1$.

1. Separate: $\dfrac{dy}{y} = x\,dx$.
2. Integrate: $\ln|y| = \dfrac{x^2}{2} + C$.
3. Solve for y: $y = Ae^{x^2/2}$.
4. Apply $y(0) = 1$: $A = 1$, so $\boxed{y = e^{x^2/2}}$.

Check: $y' = xe^{x^2/2} = xy$. Numerically, $y(1) = e^{0.5} \approx 1.648721$.

Worked example 3.2

Solve $\dfrac{dy}{dx} = \dfrac{x^2}{1+y}$, $y(0) = 0$.

1. $(1+y)\,dy = x^2\,dx$
2. $y + \dfrac{y^2}{2} = \dfrac{x^3}{3} + C$, and $y(0) = 0$ gives $C = 0$.
3. Solve the quadratic $y^2 + 2y - \frac{2}{3}x^3 = 0$ and take the root through the origin: $\boxed{y = -1 + \sqrt{1 + \frac{2}{3}x^3}}$.

At $x = 1$: $y = -1 + \sqrt{5/3} \approx 0.290994$.

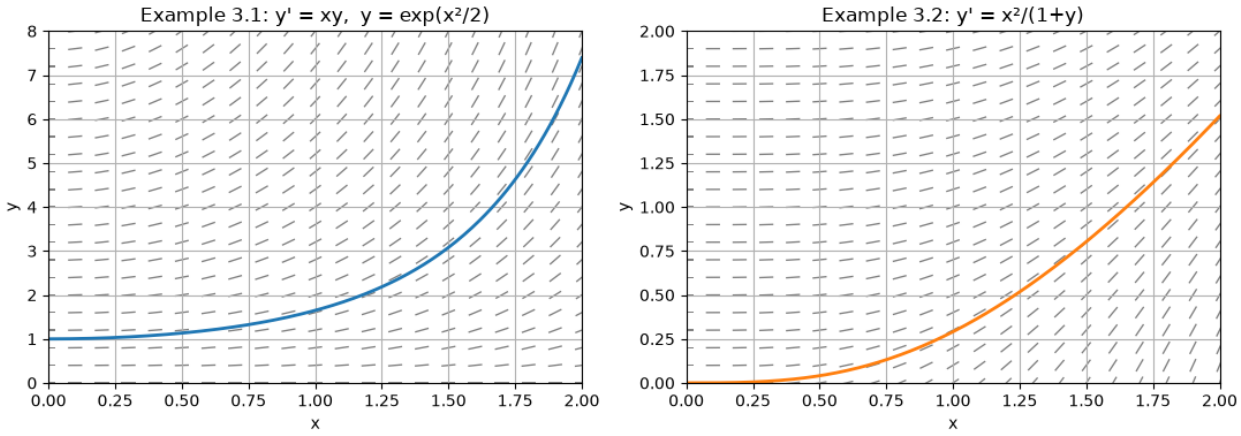
```
x = np.linspace(0, 2, 200)
fig, axes = plt.subplots(1, 2, figsize=(11, 4))

direction_field(lambda x, y: x * y, (0, 2), (0, 8), ax=axes[0])
axes[0].plot(x, np.exp(x**2 / 2), "C0", lw=2)
axes[0].set_title("Example 3.1: y' = xy,  y = exp(x²/2)")

direction_field(lambda x, y: x**2 / (1 + y), (0, 2), (0, 2), ax=axes[1])
axes[1].plot(x, -1 + np.sqrt(1 + 2 * x**3 / 3), "C1", lw=2)
axes[1].set_title("Example 3.2: y' = x²/(1+y)")
plt.tight_layout()

print("Example 3.1  y(1) =", np.exp(0.5))
print("Example 3.2  y(1) =", -1 + np.sqrt(5 / 3))
```

```
Example 3.1  y(1) = 1.6487212707001282
Example 3.2  y(1) = 0.2909944487358056
```



4. Linear first-order equations

Standard form:

$$\frac{dy}{dx} + P(x)\,y = Q(x).$$

Method (integrating factor):

1. Compute $\mu(x) = e^{\int P(x)\,dx}$.
2. Multiply through: $\mu y' + \mu P y = \mu Q$. The left side is exactly $(\mu y)'$.

3. Integrate: $\mu(x)\,y = \int \mu(x) Q(x)\,dx + C$.
4. Divide by μ.

Worked example 4.1

Solve $y' + y = x$, $y(0) = 1$.

1. $P = 1$, so $\mu = e^x$.
2. $(e^x y)' = x e^x$.
3. Integrate by parts: $e^x y = (x-1)e^x + C$.
4. $y = x - 1 + Ce^{-x}$. From $y(0) = 1$: $-1 + C = 1$, so $C = 2$.

$$\boxed{y = x - 1 + 2e^{-x}}$$

Worked example 4.2

Solve $xy' + 2y = 4x^2$, $y(1) = 2$, for $x > 0$.

1. Standard form: $y' + \dfrac{2}{x}y = 4x$, so $\mu = e^{2\ln x} = x^2$.
2. $(x^2 y)' = 4x^3$, so $x^2 y = x^4 + C$.
3. $y = x^2 + C/x^2$. From $y(1) = 2$: $C = 1$.

$$\boxed{y = x^2 + \frac{1}{x^2}}$$

```
xs = np.array([0.0, 0.5, 1.0, 1.5, 2.0])
print("Example 4.1:  y = x - 1 + 2e^(-x)")
print(f"{'x':>6} {'y(x)':>12}")
for xi in xs:
    print(f"{xi:6.2f} {xi - 1 + 2 * np.exp(-xi):12.6f}")

xs = np.array([1.0, 1.5, 2.0, 3.0])
print("\nExample 4.2:  y = x^2 + 1/x^2")
print(f"{'x':>6} {'y(x)':>12}")
for xi in xs:
    print(f"{xi:6.2f} {xi**2 + 1 / xi**2:12.6f}")
```

```
Example 4.1:  y = x - 1 + 2e^(-x)
     x         y(x)
  0.00     1.000000
  0.50     0.713061
  1.00     0.735759
  1.50     0.946260
  2.00     1.270671

Example 4.2:  y = x^2 + 1/x^2
     x         y(x)
  1.00     2.000000
  1.50     2.694444
  2.00     4.250000
  3.00     9.111111
```

5. Exact equations

An equation $M(x,y)\,dx + N(x,y)\,dy = 0$ is **exact** if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Then there is a potential function $F(x,y)$ with $F_x = M$ and $F_y = N$, and the solution is $F(x,y) = C$.

Worked example 5.1

Solve $(2xy+3)\,dx + (x^2-1)\,dy = 0$, $y(0) = 2$.

1. Test: $M_y = 2x$, $N_x = 2x$. The equation is exact.
2. Integrate M with respect to x: $F = x^2y + 3x + g(y)$.
3. Differentiate with respect to y and match N: $x^2 + g'(y) = x^2 - 1$, so $g(y) = -y$.
4. Solution: $x^2y + 3x - y = C$. From $y(0) = 2$: $C = -2$.

$$\boxed{y = \frac{-2-3x}{x^2-1} = \frac{3x+2}{1-x^2}}, \qquad -1 < x < 1.$$

At $x = 0.5$: $y = 3.5/0.75 \approx 4.666667$.

```
# Check example 5.1 by verifying that F(x, y(x)) stays equal to C = -2
y = lambda x: (3 * x + 2) / (1 - x**2)
F = lambda x, y: x**2 * y + 3 * x - y
for xi in [0.0, 0.25, 0.5, 0.75]:
    print(f"x = {xi:4.2f}   y = {y(xi):9.6f}   F(x, y) = {F(xi, y(xi)):.6f}")
```

```
x = 0.00   y =  2.000000   F(x, y) = -2.000000
x = 0.25   y =  2.933333   F(x, y) = -2.000000
x = 0.50   y =  4.666667   F(x, y) = -2.000000
x = 0.75   y =  9.714286   F(x, y) = -2.000000
```

6. Numerical methods

The methods of Sections 3–5 cover only special forms of $f(x,y)$; most equations, such as the one in Section 8, have no formula for their solution. A **numerical method** computes approximate values $y_n \approx y(x_n)$ on a grid $x_n = x_0 + nh$ with step size h. Two are used in the rest of this notebook:

Method	Formula	Order (global error)
Euler	$y_{n+1} = y_n + h\,f(x_n, y_n)$	$O(h)$
Classical Runge–Kutta (RK4)	$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ with $k_1 = f(x_n, y_n)$, $k_2 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$, $k_3 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_2)$, $k_4 = f(x_n + h, y_n + hk_3)$	$O(h^4)$

Derivations, error analysis and many more methods and examples are in the chapter **Numerical Methods for First-Order ODEs**:

1. Euler's method
2. Second-order Runge–Kutta methods
3. The classical Runge–Kutta method (RK4)

4. Error, order and efficiency
5. Stability and stiff equations
6. Adaptive step size
7. Multistep methods
8. Systems and higher-order equations

```
def euler(f, x0, y0, h, n):
    x, y = np.zeros(n + 1), np.zeros(n + 1)
    x[0], y[0] = x0, y0
    for i in range(n):
        y[i + 1] = y[i] + h * f(x[i], y[i])
        x[i + 1] = x[i] + h
    return x, y

def rk4(f, x0, y0, h, n):
    x, y = np.zeros(n + 1), np.zeros(n + 1)
    x[0], y[0] = x0, y0
    for i in range(n):
        k1 = f(x[i], y[i])
        k2 = f(x[i] + h / 2, y[i] + h / 2 * k1)
        k3 = f(x[i] + h / 2, y[i] + h / 2 * k2)
        k4 = f(x[i] + h, y[i] + h * k3)
        y[i + 1] = y[i] + h / 6 * (k1 + 2 * k2 + 2 * k3 + k4)
        x[i + 1] = x[i] + h
    return x, y
```

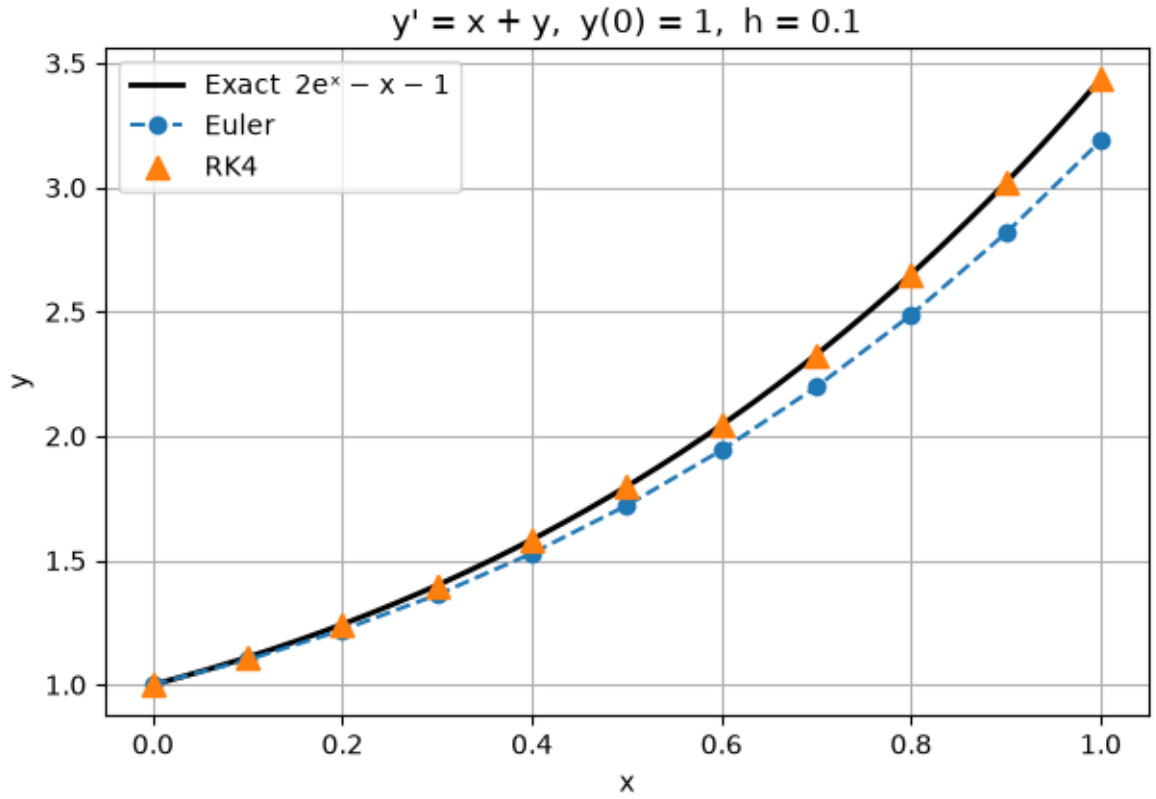
Example. $y' = x + y$, $y(0) = 1$ is linear, so the method of Section 4 gives its exact solution $y = 2e^x - x - 1$. With $h = 0.1$:

```
f = lambda x, y: x + y
exact = lambda x: 2 * np.exp(x) - x - 1

x, ye = euler(f, 0, 1, 0.1, 10)
_, yr = rk4(f, 0, 1, 0.1, 10)
print(f"y(1): exact {exact(1):.8f}   Euler {ye[-1]:.8f} (error {exact(1) - ye[-1]:.1e})   RK4 {yr[-1]:.8

xf = np.linspace(0, 1, 200)
plt.plot(xf, exact(xf), "k", lw=2, label="Exact  2e  - x - 1")
plt.plot(x, ye, "o--", label="Euler")
plt.plot(x, yr, "^", ms=8, label="RK4")
plt.title("y' = x + y,  y(0) = 1,  h = 0.1")
plt.xlabel("x"); plt.ylabel("y"); plt.legend();
```

```
y(1): exact 3.43656366   Euler 3.18748492 (error 2.5e-01)   RK4 3.43655949 (error 4.2e-
06)
```



7. Applications

7.1 Newton's law of cooling

The rate of temperature change is proportional to the difference from the surroundings:

$$\frac{dT}{dt} = -k\,(T - T_a).$$

Problem. A cup of coffee at 90°C is placed in a 20°C room with $k = 0.1$ min^{-1}. (a) What is its temperature after 10 minutes? (b) When does it reach 30°C?

The equation is separable (and linear): $T(t) = T_a + (T_0 - T_a)e^{-kt} = 20 + 70e^{-0.1t}$.

(a) $T(10) = 20 + 70e^{-1} \approx 20 + 25.752 = 45.75$°C.

(b) $20 + 70e^{-0.1t} = 30 \Rightarrow e^{-0.1t} = 1/7 \Rightarrow t = 10\ln 7 \approx 19.46$ min.

```
k, Ta, T0 = 0.1, 20.0, 90.0
cool = lambda t, T: -k * (T - Ta)
T_exact = lambda t: Ta + (T0 - Ta) * np.exp(-k * t)

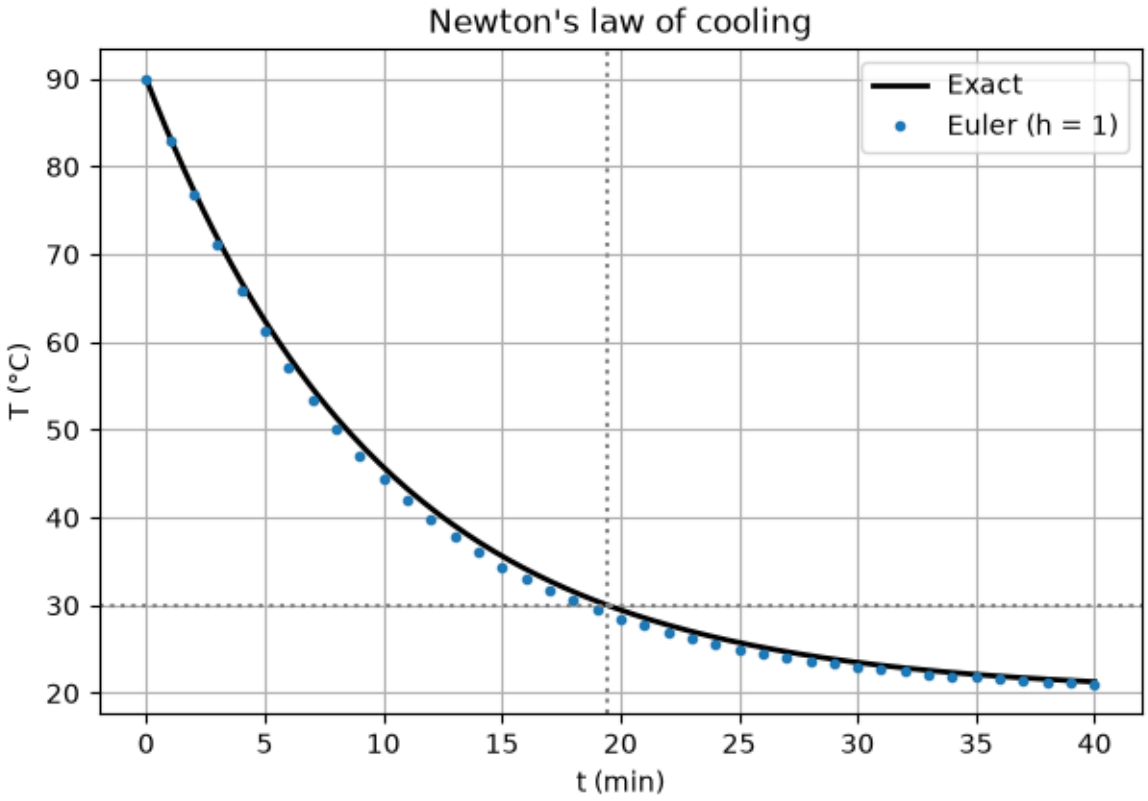
t, T_rk = rk4(cool, 0, T0, 1.0, 40)       # h = 1 minute
_, T_eu = euler(cool, 0, T0, 1.0, 40)

print(f"T(10) exact = {T_exact(10):.4f}   RK4 = {T_rk[10]:.4f}   Euler = {T_eu[10]:.4f}")
print(f"time to reach 30 °C = 10 ln 7 = {10 * np.log(7):.4f} min")

tf = np.linspace(0, 40, 200)
plt.plot(tf, T_exact(tf), "k", lw=2, label="Exact")
```

```
plt.plot(t, T_eu, ".", label="Euler (h = 1)")
plt.axhline(30, color="gray", ls=":"); plt.axvline(10 * np.log(7), color="gray", ls=":")
plt.xlabel("t (min)"); plt.ylabel("T (°C)"); plt.title("Newton's law of cooling")
plt.legend();
```

```
T(10) exact = 45.7516   RK4 = 45.7516   Euler = 44.4075
time to reach 30 °C = 10 ln 7 = 19.4591 min
```



7.2 Logistic population growth

$$\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right),$$

where r is the growth rate and K is the carrying capacity. Separating variables with partial fractions gives

$$P(t) = \frac{K}{1 + \left(\dfrac{K - P_0}{P_0}\right) e^{-rt}}.$$

Problem. $r = 0.5$ per year, $K = 1000$, $P_0 = 100$. Find $P(5)$.

$$P(5) = \frac{1000}{1 + 9e^{-2.5}} = \frac{1000}{1 + 9(0.082085)} = \frac{1000}{1.738765} \approx 575.1.$$

The equilibria are $P = 0$ (unstable) and $P = K$ (stable). Growth is fastest at $P = K/2$.

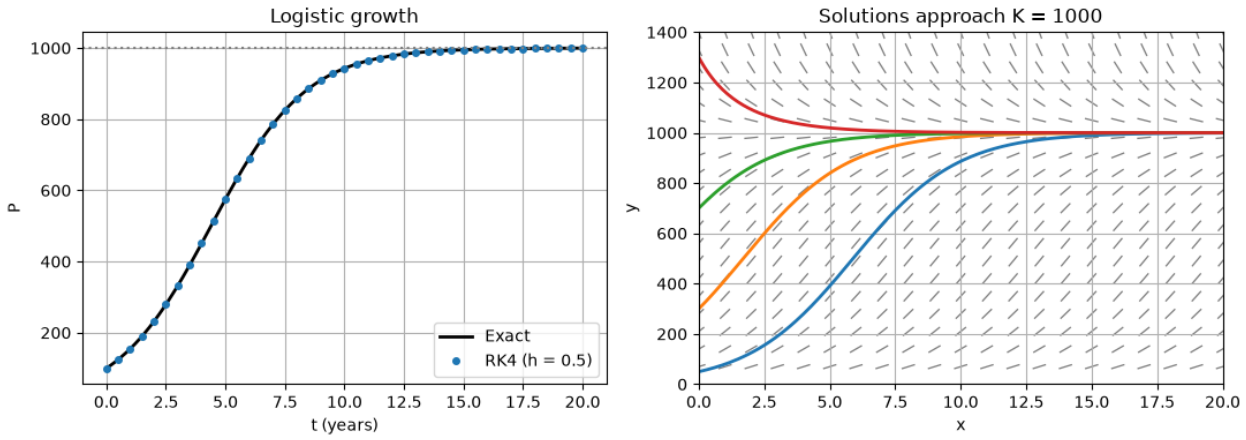
```
r, K, P0 = 0.5, 1000.0, 100.0
logistic = lambda t, P: r * P * (1 - P / K)
P_exact = lambda t: K / (1 + (K - P0) / P0 * np.exp(-r * t))

t, P_rk = rk4(logistic, 0, P0, 0.5, 40)
print(f"P(5) exact = {P_exact(5):.4f}   RK4 (h=0.5) = {P_rk[10]:.4f}")

fig, axes = plt.subplots(1, 2, figsize=(11, 4))
tf = np.linspace(0, 20, 200)
axes[0].plot(tf, P_exact(tf), "k", lw=2, label="Exact")
axes[0].plot(t, P_rk, "o", ms=4, label="RK4 (h = 0.5)")
axes[0].axhline(K, color="gray", ls=":"); axes[0].set_title("Logistic growth")
axes[0].set_xlabel("t (years)"); axes[0].set_ylabel("P"); axes[0].legend()

direction_field(logistic, (0, 20), (0, 1400), ax=axes[1])
for p0 in [50, 300, 700, 1300]:
    _, Pp = rk4(logistic, 0, p0, 0.1, 200)
    axes[1].plot(np.linspace(0, 20, 201), Pp, lw=2)
axes[1].set_title("Solutions approach K = 1000")
plt.tight_layout()
```

```
P(5) exact = 575.1209   RK4 (h=0.5) = 575.1165
```



8. An equation with no elementary solution

The Riccati equation

$$y' = x^2 + y^2, \qquad y(0) = 0$$

cannot be solved in terms of elementary functions. We estimate $y(1)$ with RK4 and check the result by repeatedly halving h. When the digits stop changing, the answer has converged.

```
g = lambda x, y: x**2 + y**2

prev = None
print(f"{'h':>8} {'y(1) RK4':>16} {'change':>12}")
for h in [0.2, 0.1, 0.05, 0.025, 0.0125]:
    _, y = rk4(g, 0, 0, h, round(1 / h))
    change = f"{abs(y[-1] - prev):12.2e}" if prev is not None else ""
```

```
    print(f"{h:8.4f} {y[-1]:16.10f} {change}")
    prev = y[-1]
```

```
       h          y(1) RK4        change
  0.2000     0.3502575491
  0.1000     0.3502337418     2.38e-05
  0.0500     0.3502319725     1.77e-06
  0.0250     0.3502318526     1.20e-07
  0.0125     0.3502318448     7.79e-09
```

The last two values agree to about 8 significant figures: $y(1) \approx 0.3502318$. The solution grows quickly afterwards and blows up near $x \approx 2.003$, a reminder that solutions of nonlinear ODEs can exist only on a finite interval.

9. Summary

Type	Form	Method
Separable	$y' = g(x)h(y)$	$\int dy/h = \int g\,dx$
Linear	$y' + Py = Q$	Integrating factor $\mu = e^{\int P}$
Exact	$M\,dx + N\,dy = 0$, $M_y = N_x$	Find F with $F_x = M$, $F_y = N$; then $F = C$
Anything else	$y' = f(x, y)$	Numerical: Euler (order 1), RK4 (order 4); see Numerical Methods

10. Exercises

1. Solve $y' = y\cos x$, $y(0) = 2$.
2. Solve $y' - 3y = e^{2x}$, $y(0) = 0$.
3. Show that $(ye^x + 2x)\,dx + (e^x - 1)\,dy = 0$ is exact and solve it with $y(1) = 0$.
4. Use Euler with $h = 0.25$ to estimate $y(1)$ for $y' = -2xy$, $y(0) = 1$. Compare with the exact value e^{-1}.
5. Repeat Exercise 4 with RK4 and $h = 0.25$. How many times smaller is the error?

Answers

1. $y = 2e^{\sin x}$
2. $y = e^{3x} - e^{2x}$
3. $M_y = e^x = N_x$; $F = ye^x + x^2 - y$, so $ye^x + x^2 - y = 1$, giving $y = \dfrac{1 - x^2}{e^x - 1}$.
4. and 5. Run the cell below.

```
q = lambda x, y: -2 * x * y
_, y_eu = euler(q, 0, 1, 0.25, 4)
_, y_rk = rk4(q, 0, 1, 0.25, 4)
ex = np.exp(-1)
print(f"Exact  y(1) = {ex:.6f}")
print(f"Euler  y(1) = {y_eu[-1]:.6f}   error = {abs(y_eu[-1] - ex):.2e}")
print(f"RK4    y(1) = {y_rk[-1]:.6f}   error = {abs(y_rk[-1] - ex):.2e}")
print(f"Error ratio Euler / RK4   {abs(y_eu[-1] - ex) / abs(y_rk[-1] - ex):.0f}")
```

```
Exact  y(1) = 0.367879
Euler  y(1) = 0.410156   error = 4.23e-02
RK4    y(1) = 0.367934   error = 5.51e-05
Error ratio Euler / RK4   768
```